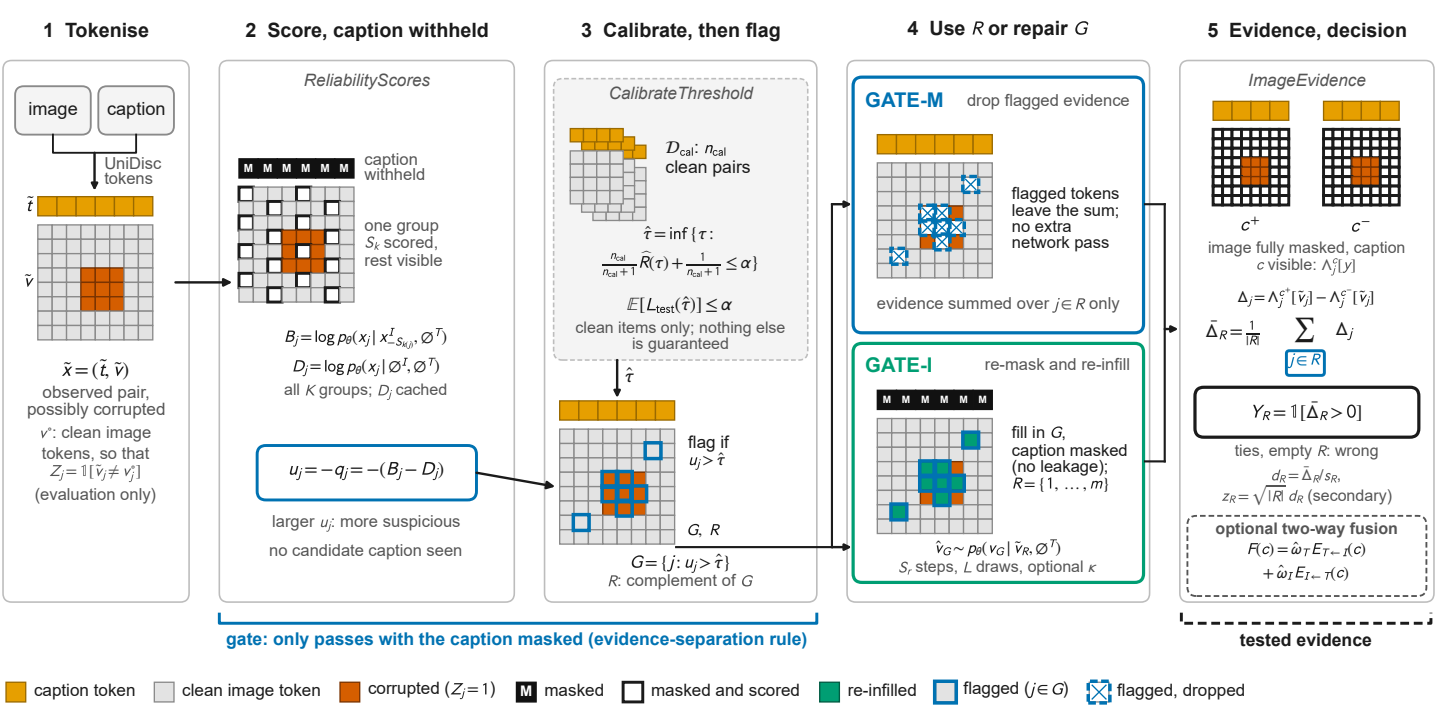

1 Tokenise
2 Score, caption withheld
3 Calibrate, then flag
4 Use R or repair G
5 Evidence, decision
image
caption
UniDisc tokens
$\tilde{t}$
$\tilde{v}$
$\tilde{x} = (\tilde{t}, \tilde{v})$
observed pair, possibly corrupted
v^*: clean image tokens, so that $Z_j = \mathbb{1}[\tilde{v}_j \neq v_j^*]$
(evaluation only)
ReliabilityScores
caption withheld
one group S_k scored, rest visible
$B_j = \log p_\theta(x_j \mid x^{\mathcal{I}}_{S_{k(j)}}, \varnothing^{\mathcal{T}})$
$D_j = \log p_\theta(x_j \mid \varnothing^{\mathcal{I}}, \varnothing^{\mathcal{T}})$
all K groups; D_j cached
$u_j = -q_j = -(B_j - D_j)$
larger u_j: more suspicious
no candidate caption seen
CalibrateThreshold
$\mathcal{D}_{\text{cal}}$: n_{cal} clean pairs
$\hat{\tau} = \inf\{\tau : \frac{n_{\text{cal}}}{n_{\text{cal}}+1}\hat{R}(\tau) + \frac{1}{n_{\text{cal}}+1} \le \alpha\}$
$\mathbb{E}[L_{\text{test}}(\hat{\tau})] \le \alpha$
clean items only; nothing else is guaranteed
$\hat{\tau}$
flag if $u_j > \hat{\tau}$
G, R
$G = \{j : u_j > \hat{\tau}\}$
R: complement of G
GATE-M
drop flagged evidence
flagged tokens leave the sum; no extra network pass
evidence summed over $j \in R$ only
GATE-I
re-mask and re-infill
fill in G, caption masked (no leakage); $R = \{1, \ldots, m\}$
$\hat{v}_G \sim p_\theta(v_G \mid \tilde{v}_R, \varnothing^{\mathcal{T}})$
S, steps, L draws, optional κ
ImageEvidence
c^+
c^-
image fully masked, caption c visible: $\Lambda_j^c[y]$
$\Delta_j = \Lambda_j^{c^+}[\tilde{v}_j] - \Lambda_j^{c^-}[\tilde{v}_j]$
$\bar{\Delta}_R = \frac{1}{|R|} \sum_{j \in R} \Delta_j$
$Y_R = \mathbb{1}[\bar{\Delta}_R > 0]$
ties, empty R: wrong
$d_R = \bar{\Delta}_R / s_R$,
$z_R = \sqrt{|R|}\, d_R$ (secondary)
optional two-way fusion
$F(c) = \hat{\omega}_T E_{T \leftarrow \mathcal{I}}(c)$
$+ \hat{\omega}_I E_{\mathcal{I} \leftarrow \mathcal{T}}(c)$
gate: only passes with the caption masked (evidence-separation rule)
tested evidence
caption token
clean image token
corrupted ($Z_j = 1$)
masked
masked and scored
re-infilled
flagged ($j \in G$)
flagged, dropped